Project Initialization and Planning Phase

Date	28 October 2026
Team ID	EN25CS3010655
Project Title	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build Health360, an AI-driven and IoT-integrated healthcare analytics platform that continuously monitors patient vitals, performs multi-disease risk prediction, and alerts medical professionals to enable early, proactive interventions.
Scope	Includes real-time data ingestion from wearable sensors, machine learning models for early disease detection (cardiovascular, diabetes, respiratory), interactive web/mobile dashboards for patients and doctors, and HIPAA/GDPR-compliant encrypted cloud storage. Excludes direct surgical/robotic interventions.
Problem Statement	
Description	Traditional healthcare systems are predominantly reactive, fragmented, and burdened by data silos. Patients with chronic conditions lack continuous monitoring between clinic visits, leading to delayed diagnoses, preventable complications, and high hospital readmission rates.
Impact	Solving this problem transitions care from reactive to preventive, significantly reducing critical medical emergencies, lowering healthcare costs for patients, minimizing clinical workload, and improving overall survival and quality of life.
Proposed Solution	

Approach	Health360 employs a 5-layer IoT-Cloud-AI architecture: 1) Non-invasive wearable sensing, 2) Encrypted MQTT/REST data transmission, 3) Automated data cleaning and ETL pipelines, 4) Predictive analytics via supervised ML models (Scikit-Learn/TensorFlow), and 5) Real-time alert notifications and interactive React web/mobile dashboards.
Key Features	Continuous vital sign tracking (Heart Rate, SpO2, Temperature); Automated ML-based disease risk scoring; Real-time emergency alert triggers for doctors and caregivers; Role-based dashboards for patients, doctors, and admins; Secure, encrypted EHR data management.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU / GPU specs for model training & API service	8-Core Intel/AMD CPU, 1 x NVIDIA RTX 3080 / T4 GPU
Memory	System RAM for pipeline processing & data serving	16 GB DDR4 RAM
Storage	Cloud / Local SSD for EHR storage, ML models, and logs	512 GB NVMe SSD + AWS S3 Storage
Software		
Frameworks	Backend REST API & Web Frameworks	Flask / FastAPI, React.js
Libraries	Data analytics, machine learning, and visualization	scikit-learn, pandas, numpy, TensorFlow, Chart.js
Development Environment	IDE, testing environment, and version control	VS Code, Jupyter Notebook, Git, GitHub, Postman
Data		
Data	Synthetic & public medical datasets (EHR, vitals, risk indicators)	MIMIC-III / Kaggle Healthcare Datasets (~50,000 anonymized patient records, CSV format)